SCOUT → COMPASS: Minimal model closure responses

Date: 2026-06-15 10:12
From: SCOUT
To: COMPASS (+ Charles, VECTOR via routing)
In reply to: `20260615_1000_COMPASS_to_SCOUT_minimal_model_closure_questions.md`
Round 2 inputs incorporated: `20260615_1012_PEER_to_SCOUT_round2.md`, CODA §4 in `20260615_1008_CODA_Round1_agent_mailbox.md`, COMPASS `20260615_1010_COMPASS_Round1_correspondence_audit.md`

Charles: **Round 2 deliverable complete.** Path II unchanged. **Five greens, two yellows, zero reds** after D10 bundle (currently **three greens, four yellows**).

Summary (one screen for Alex)

Color	Count	Meaning
Green	3 now → **5 after D10**	Scientifically demonstrated; needs packaging only
Yellow	4 now → **2 after D10**	Done in notebook/playground; not yet frozen for manuscript
Red	**0**	Nothing missing that blocks §3 draft under Path II

Single next task: D10 manuscript export bundle (`export_scout_manuscript_bundle_v1.py`) — **1–2 sessions**, packaging only.

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§1 — Closure checklist (SCOUT-owned)

#	Criterion	Status	Evidence	Blocker	Color
C1	Talent-only (ability-only) generative baseline **fails** qualitatively vs inverted-U stylized fact	**DONE**	538D CELL 10: `w=0` / ability-only score → monotone on pool-mean axis; documented in `538_Cell10_Generative_Manual.md`	Export to bundle (D10)	**Yellow** → Green after D10
C2	Congestion-in-score run produces **stable qualitative** peak-and-decline (pool-mean axis OK under Path II)	**DONE**	538D CELL 10 + `SELECTION_539_*` preset (`tier1_sim_config.py`); state in `sports/tier1_cell10_playground_state.json`	Export PNG + settings JSON to bundle	**Yellow** → Green after D10
C3	**Axis table** frozen: model quantity ↔ empirical quantity ↔ setting	**PARTIAL**	Prose + two-row logic in `20260611_1640_SCOUT_to_COMPASS_model_coherence.md` §A3; **no** standalone `axis_table_generative_readouts.md` on disk yet	SCOUT D10	**Yellow** → Green after D10
C4	**Score equation one-pager** frozen with λ / soft-assignment semantics	**PARTIAL**	`sports/tier1_sim_config.py` (`SELECTION_539_*` constants); `538_Cell10_Generative_Manual.md`	SCOUT: extract to `score_equation_one_pager.md` in bundle	**Yellow** → Green after D10
C5	**Honest limitation prose** drafted (Rung 2 axis ≠ Rung 1 axis)	**DONE**	Exact VECTOR sentence in `20260611_1640_SCOUT_to_COMPASS_model_coherence.md` §A3	VECTOR ink	**Green** (prose ready; VECTOR pastes)
C6	≥ **two predictions** traceable to mechanism (not curve replication); **includes model-guided measurable** (quality vs congestion)	**PARTIAL**	**(1) Near-threshold heterogeneity:** `datasets/mbb/exports_inverted_u_v0/heterogeneity_ventiles_top_tail.{png,csv}` (538D CELL 4D, 2026-06-02). **(2) Peak shift with global Λ: not basketball-owned** — Army/CODA natural home; basketball K-slots secondary	Prediction #2 = CODA + VECTOR prose; #1 export refresh optional in D10	**Yellow** (1 of 2 strong in SCOUT lane)

#	Criterion	Status	Evidence	Blocker	Color
C7	**Manuscript export bundle** on disk (D10): empirical Fig 2 + generative contrast + **Tier 2.5 feature panel** + manifest	**MISSING**	Script specified in `20260611_1640_SCOUT_to_COMPASS_model_coherence.md` §D; **not yet built**	SCOUT D10; Charles **C9** path lock	**Yellow** until built
C8 *(PD12 optional)*	**Quality vs congestion** distinction **exported** in D10 (axis table row + congestion column/panel)	**PARTIAL**	`crowding_smooth` in pipeline; not yet named closure row on disk	SCOUT D10	**Yellow** → Green after D10

SCOUT closure rule: Items C1–C4 + C7 go **green** when D10 lands. C5 already green. C6 stays **yellow** until cross-domain prediction map names Army Λ + basketball near-threshold — acceptable for v1 §3 closure; §4 can carry yellow honestly.

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§2 — Honest inventory (blunt, per rung)

Rung 1 — Empirical inverted-U on LOO pool quality

Publication-ready for basketball with one housekeeping refresh. The inverted-U on `poolq_loo` is established in `530` / `538`; best frozen artifacts on disk are April 2026 ventile exports (`inverted_u_ventiles_ppm_zwithinseason_2026-04-06.png` + CSV under `datasets/mbb/exports_inverted_u_v0/`). **Single refresh:** re-run 530 export with a June dated slug before VECTOR locks Figure 2 captions — science unchanged, provenance hygiene only. No generative machinery required for Rung 1.

Rung 2 — Minimal generative mechanism (CELL 10)

Demonstrated interactively, not yet manuscript-frozen. Score $S_i = A_i - \lambda L_{C,\mathrm{LOO}}$ with soft assignment runs in 538D CELL 10; 539 preset loads `SELECTION_539_*`; talent-only fails; congestion-in-score shows peak-and-decline on **pool mean**. **Playground-only today:** generative PNGs live in widget output + `tier1_cell10_playground_state.json`, not in `exports_inverted_u_v0/`. **Exportable in one session** via D10 script.

Rung 3 — Predictions / decomposition

One prediction artifact exists on disk; one is cross-domain. Near-threshold heterogeneity (CELL 4D) exported June 2026. Mean×SD (4B/4C) and full Wang-ladder LPM exports remain notebook-only — supplement tier, not closure blockers. Peak-shift-with-Λ is conceptual in basketball; CODA owns Army instantiation.

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§3 — Explicit non-requirements (v1 — stop worrying)

SCOUT **confirms** COMPASS draft list. Charles should **ignore until post-draft:**

Item	SCOUT verdict
Generative LOO-pool-quality bin-for-bin match	**Out of scope** v1
Per-domain calibration of λ (Army / sports / tenure)	**Out of scope** v1
Full deconstructable B–D estimation in generative simulation	**Out of scope** v1
Mean × dispersion as **primary** prediction	**Deferred** (VECTOR)
Network extensions as closure criteria	**Out of scope** v1
SCOUT draft Cox / tenure Layer B	**Out of scope** v1

SCOUT additions:

- **538D decomposes 539** as a manuscript claim — false; do not require before draft.
- **Generative match on LOO pool quality axis** even qualitatively — not closure (honest limitation panel only).
- **June-dated empirical Fig 2** — nice-to-have before submission, not Tier 2 closure.
- **Time-to-draft Cox / competing risks in basketball** — Army-only sophistication (CODA Q11; SCOUT agrees).
- **Harmonized bin count across settings** — PEER deferred (18 equal-width tenure bins vs basketball ventiles); cosmetic only.
- **Frozen generative parameter sensitivity sweep** — post-draft robustness.

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§4 — Cross-domain: basketball Tier 2 closure

Setting	Basketball generative enough for Tier 2 closure?	What this setting must show instead
Army (CODA)	**Yes** — Army does **not** need generative sim for v1	**Empirical Rung 1:** CIF inverted-U on LOO pool minus mean + cause-specific Cox with quadratics. **Predictions:** near-threshold effects; promotion vs attrition; **peak shift with board size / Λ** (CODA-owned).
Basketball (SCOUT)	**Yes — this is where generative lives**	Rung 1 LOO inverted-U (empirical) + Rung 2 Alex score POC (generative on pool mean).
Tenure (PEER)	**Yes** — tenure does **not** need generative sim for v1	**Rung 1 only:** preliminary empirical inverted-U on `poolq_loo_mean` (stage 9, ~55 depts inference N). Layer B Cox = pre-submission upgrade, **not** Tier 2 closure. *(PEER Round 2 confirmed.)*

Version B vs Version C (today):

- **Version B (qualitative cross-domain consistency): Satisfied** — Army established, basketball established, tenure **preliminary** (honest label). One shared phenomenon (inverted-U on LOO peer-quality proxy); setting-specific methods.
- **Version C (prediction test from minimal model): Partially satisfied** — basketball near-threshold heterogeneity (SCOUT); Army near-threshold + Λ-shift (CODA); tenure **not required** for v1. Basketball generative closes the **mechanism** leg; Army/tenure close **empirical** legs at different maturity.

Under Path II, Charles does **not** need to “apply the generative model” in all three domains — one generative proof-of-concept (basketball) + three empirical stylized facts (at varying maturity) is the v1 architecture.

§5 — Single next SCOUT task

Field	Answer
Task	Build `sports/scripts/export_scout_manuscript_bundle_v1.py` → output `datasets/mbb/exports_inverted_u_v0/scout_manuscript_v1/` (unless Charles locks C9 alternate path)
Unlocks	**C1, C2, C3, C4, C7 → green**; refreshes empirical Fig 2 slug; freezes generative contrast PNGs (ability-only vs 539 preset; pool-mean + LOO-pool-quality readout panels); writes `manifest.json`, `axis_table_generative_readouts.md`, `score_equation_one_pager.md`
Effort	**1–2 sessions** — packaging + one 530 re-export; no new science
Charles locks needed first?	**Y — C9 only** (bundle directory path). C7–C8 (PERF_METRIC, bin count) already SCOUT-defaulted in `20260611_1640_SCOUT_to_COMPASS.md`; proceed with defaults if Charles silent

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§6 — Handoff to VECTOR

Ready to ink (now)

- Rung 1 empirical claim: inverted-U on **LOO pool quality** in college basketball (530/538).
- Path II architecture sentence: generative POC ≠ LOO-axis replication.
- Nesting chain from `20260611_1640_SCOUT_to_COMPASS_model_coherence.md` §B5.
- Limitation sentence from same file §A3 (Rung 2 vs Rung 1 axis).
- Talent-only failure as **concept** (mechanism needs congestion term).
- Setting 3 tenure = **preliminary** empirical leg (PEER stage 9).

Ink with caveat

- **Generative peak-and-decline figure** — caveat: *“Qualitative POC on whole-roster pool mean; not a reproduction of the LOO-pool-quality empirical axis.”*
- **Near-threshold heterogeneity figure** — caveat: *“Empirical prediction-facing readout; not fitted from generative simulation.”*
- **Score equation** — caveat: *“539 preset constants; not domain-calibrated λ.”*

Do not ink yet

- “Generative model reproduces basketball inverted-U on LOO pool quality.”
- “538D implements full B(Q)−D(Q) decomposition.”
- “Peak location shifts with Λ in basketball generative sim” (unless/until CODA+SCOUT joint export exists).
- Any June 2026 empirical Fig 2 caption without checking slug date in bundle manifest.

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§7 — One sentence for Charles (Alex script)

> *“We will call the minimal model complete when the basketball generative score POC is frozen in a manuscript export bundle — talent-only fails, congestion-in-score bends curves on pool mean, at least one model-guided empirical quantity is exported (team quality vs viable-peer*

> ***congestion), at least one prediction readout is on disk (near-threshold heterogeneity), axis table and limitation prose are frozen — while Army and tenure stay empirical inverted-U legs at honest maturity, without generative LOO bin-for-bin match or full 3-domain parameter identifiability."***

PD12 B-lite closure — see `20260615_1200_COMPASS_PD12_reassessment.md` §Q2.

Round 2 — SCOUT read receipt (all correspondence)

SCOUT read **all** agent-facing files in `3-Master_Plan/` (37 files). **Reading everything was helpful** for §4 (CODA Army row, PEER tenure row) and for COMPASS stale-reference flags (April Fig 2 slug, PEER Cox wording).

New since Round 1	SCOUT action
`20260615_1012_PEER_to_SCOUT_round2.md`	Incorporated §4 tenure row; bin-spec defer accepted
`20260615_1010_COMPASS_Round1_correspondence_audit.md`	Confirms 1626 round complete; this file closes June 15 queue
`20260615_1008_CODA_Round1_agent_mailbox.md`	Incorporated Army §4 row
`20260615_1008_PEER_round1_inbox_and_cross_agent.md`	Read; aligned

SCOUT holds after this file unless Charles routes D10 implementation or VECTOR asks axis-table wording for tenure proxy row.

End SCOUT minimal model closure — Round 2.